

1. Administrative & Study Identification

Full Study Title: A Prospective, Open-Label, Single-Arm, Phase II Study of Acalabrutinib and Rituximab in Untreated Elderly and/or Frail Patients With Diffuse Large B-Cell Lymphoma (ACRUE) **Short Title/Acronym:** Study of Acalabrutinib and Rituximab in Untreated Elderly and/or Frail Patients With DLBCL (ACRUE) **Protocol Number:** D8227C00002 **NCT Number:** NCT05952024 **Posting ID:** 76468885591484880 **Sponsor Name:** AstraZeneca **Company:** AstraZeneca **Trial Status:** Recruiting **Investigational Product:** Acalabrutinib and Rituximab (ATC Code: L01FA01) **Phase of Development:** Phase II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of acalabrutinib in combination with rituximab by measuring the percentage of patients experiencing Grade 3-4 treatment-emergent adverse events (TEAEs). **Secondary Objectives:** To evaluate the clinical efficacy of this combination regimen, including objective response rate (ORR), progression-free survival (PFS), and event-free survival (EFS) in this patient population.

2.2 Background & Rationale Patients aged ≥ 80 years or frail patients with diffuse large B-cell lymphoma (DLBCL) are often unable to tolerate standard-of-care chemoimmunotherapy (CIT) regimens like R-CHOP or R-miniCHOP due to high treatment-related mortality and toxicity rates. The ACRUE study is designed to address this critical unmet medical need by providing a targeted, chemo-free treatment alternative. Acalabrutinib, a highly selective Bruton tyrosine kinase (BTK) inhibitor, combined with rituximab, aims to offer a safe, effective, and tolerable frontline therapeutic option for this vulnerable patient group.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 80 patients **Number of Sites:** Global, multicenter **Estimated Study Start:** 2023 **Estimated Primary Completion:** July 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. ≥ 80 years of age at the time of screening, OR ≥ 65 to 79 years of age at the time of screening and considered ineligible for chemoimmunotherapy. 2. Histologically documented DLBCL. 3. No prior treatment for DLBCL. 4. Stage II, III, or IV disease by the Ann Arbor Classification. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2 with no deterioration over the previous 2 weeks prior to baseline or day of the first dosing except when due to underlying lymphoma. 6. At least 1 lesion that can be accurately measured at baseline as ≥ 10 mm in the longest diameter with computed tomography or magnetic resonance imaging and is suitable for accurate repeated measurements. 7. Adequate organ and marrow function independent of growth factor or transfusion support within 1 week of Screening.

4.2 Exclusion Criteria 1. Any evidence of diseases (such as severe or uncontrolled systemic diseases, including uncontrolled hypertension, renal transplant, and active bleeding diseases) that would make the study undesirable for the patient or that would impact compliance with the protocol. 2. History of prior or current malignancy that would affect compliance with the protocol or interpretation of the results. 3. Serologic status reflecting active hepatitis B or C infection. 4. Serological positivity or known infection with HIV. 5. Active central nervous system involvement by lymphoma, leptomeningeal disease, or spinal cord compression. 6. Any comorbidity or organ system impairment rated with a single Cumulative Illness Rating Scale-Geriatric score (CIRS-G) of 4 or a total CIRS-G score > 17 . 7. History of or ongoing confirmed Progressive Multifocal Leukoencephalopathy. 8. Known active significant infection. 9. History of stroke or intracranial haemorrhage within 6 months before the first dose of study drug. 10. History of bleeding diathesis (eg, haemophilia, von Willebrand disease). 11. Major surgical procedure within 30 days of first dose of study intervention or anticipated major surgery during the study timeframe. 12. Requires or is receiving anticoagulation with warfarin or equivalent vitamin K antagonists. 13. Received a live virus vaccination within 28 days of the first dose of study drug.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Acalabrutinib:** 100 mg orally twice daily beginning on Day 1 of Cycles 1-28. * **Rituximab:** 375 mg/m² intravenously on Cycle 1 Day 15; followed by 1400 mg subcutaneously on Day 1 of Cycles 2-8.

Route of Administration: Oral (Acalabrutinib), Intravenous and Subcutaneous (Rituximab). **Duration of Treatment:** Up to 104 weeks of treatment and follow-up. Rituximab is given for up to 8 cycles, and acalabrutinib for up to 28 cycles (Cycles 1-8 are 21

days; Cycles 9-28 are 28 days). The total study duration per patient will be up to 108 weeks, including up to 28 days for screening.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive therapies for co-morbidities not interfering with BTK inhibitors. **Prohibited:** Warfarin or equivalent vitamin K antagonists, and administration of a live virus vaccination within 28 days prior to the first dose of the study drug.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, CIRS-G assessment, Imaging (CT/MRI)
Baseline	C1 Day 1	Treatment initiation (Acalabrutinib), Safety Labs
Interim	Cycles 1-8	Rituximab administration (IV C1D15, SC C2-C8 Day 1), toxicity monitoring, Efficacy Assessments
Maintenance	Cycles 9-28	Acalabrutinib monotherapy, regular safety monitoring, ongoing response assessment
End of Study	Week 108	Final safety follow-up, disease evaluation, treatment discontinuation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of patients with Grade 3 to 4 treatment emergent adverse events (TEAEs).

Measurement Method: Clinical safety evaluation per standard adverse event reporting criteria over the 108-week study duration.

7.2 Secondary Endpoints * Objective Response Rate (ORR) * Progression-Free Survival (PFS) * Event-Free Survival (EFS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at every cycle visit with a primary focus on Grade 3-4 TEAEs, particularly bleeding events and infections common with BTK inhibitors.

Serious Adverse Events (SAEs): Routine reporting to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal and external safety review committees managed by AstraZeneca.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for the primary endpoint evaluation (toxicity-based); Intent-to-Treat (ITT) for overall

efficacy signals. **Sample Size Justification:** A target enrollment of $N=80$ is scaled to appropriately characterize the safety profile and incidence of Grade 3-4 TEAEs in this specific frail/elderly cohort. **Handling of Missing Data:** Best-practice conventions including Last Observation Carried Forward (LOCF) for relevant clinical variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to ensure protocol adherence, especially regarding frailty/CIRS-G scoring and adverse event tracking. **Audit Trail:** Formalized data tracking via eCRF and routine clinical quality audits.

11. Study Identifiers & Governance

Primary ID: D8227C00002 **Registry Number:** NCT05952024 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Study of Acalabrutinib and Rituximab in Untreated Elderly and/or Frail Patients With DLBCL **Official Regulatory Title:** A Prospective, Open-Label, Single-Arm, Phase II Study of Acalabrutinib and Rituximab in Untreated Elderly and/or Frail Patients With Diffuse Large B-Cell Lymphoma

12. Clinical Framework (DLBCL Specific)

Disease Area: Diffuse Large B-Cell Lymphoma (DLBCL) **Primary**

Condition: Diffuse Large B-Cell Lymphoma (DLBCL) **Diagnosis**

Threshold: Histologically documented untreated DLBCL, Stage II-IV

Medical Methodology: Targeted BTK inhibitor + Monoclonal Antibody (Chemo-free regimen) **Optimization Target:** Providing an effective therapy while minimizing severe toxicity and avoiding standard chemoimmunotherapy in frail/elderly populations

13. Study Procedures & Methodology

Management Pathway: Oral Acalabrutinib combined with SC/IV Rituximab induction followed by Acalabrutinib maintenance.

Education Protocol (HCP): Site training on appropriate CIRS-G frailty scoring and the management of BTK inhibitor specific side-effects.

Education Protocol (Patient): Strict guidance on twice-daily oral medication adherence and recognizing signs of bleeding/infection.

Quality Audit Mechanism: Periodic review of enrollment criteria strictly focusing on the elderly/frail population definitions.

14. Observation & Follow-Up Schedule

Total Duration: Up to 108 weeks (104 weeks of active treatment/follow-up) **Onsite Visit Cadence:** Day 1 of each 21-day (Cycles 1-8) and 28-day (Cycles 9-28) cycle **Remote Monitoring Frequency:** Every 4 weeks (Q4W) post-induction phase if applicable. **Checklist Review Interval:** At the beginning of each cycle for drug accountability and safety.

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					